General-Sum Games and Correlated Equilibrium

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General-Sum Games

Normal-Form Game

$$[M], (\mathcal{A}_j)_{j\in[M]}, (u_j)_{j\in[M]}$$

- Player $1, 2, \dots, M$
- Player j choose action $a_j \in \mathcal{A}_j$
- Player j's payoff/utility is $u_j(a_1, \dots, a_M) = u_j(a_j, a_{-j})$

$$a_{-j} \coloneqq (a_1, a_2, \dots, a_{j-1}, a_{j+1}, \dots, a_M)$$

Two-player zero-sum game is a special case when $M = 2$ and $u_1 = -u_2$

Example: Prisoner's Dilemma

	Cooperate	Defect
Cooperate	-1,-1	-10,0
Defect	0,-10	-6,-6

Prisoner's Dilemma

	High	Low
High	5,5	0,6
Low	6,0	3,3

Pricing game

General form:

	C	D
C	x,x	z,w
D	w,z	y,y

w > x > y > z

Example: Chicken Game

	Chicken	Dare
Chicken	6,6	2,7
Dare	7,2	0,0

Chicken game

	Compromise	Stand firm
Compromise	0,0	-5,5
Stand firm	5,-5	-10,-10

Military confrontation / negotiation

General form:

	C	D
C	x,x	z,w
D	w,z	y,y

w > x > z > y

Example: Stag Hunt

	Stag (C)	Hare (D)
Stag (C)	6,6	2,7
Hare (D)	7,2	4,4

Stag Hunt

	Wait	Withdraw
Wait	5,5	0,4
Withdraw	4,0	2,2

Bank runs

General form:

	C	D
C	x,x	z,w
D	w,z	y,y

x > w > y > z

Equilibrium

- An equilibrium is a **steady state** where no player wants to unilaterally deviate
  - Different strategy spaces and allowed deviations give different equilibrium concepts

- Traditional game theory studies equilibria. Studying equilibria is meaningful because it allows a player to
  - Predict other players' moves
  - Reason about their own move

  It also allows the game designer (e.g. government) to
  - Design the rules of the game so that the equilibrium outcome is desirable **(mechanism design)**

Nash Equilibrium (NE)

$(P_1, P_2, \cdots, P_M) \in \Delta_{A_1} \times \Delta_{A_2} \times \cdots \Delta_{A_N}$ is called a NE if

$$u_j(P_j, P_{-j}) \geq u_j(P_j', P_{-j}) \qquad \forall P_j' \in \Delta_{A_j}$$

$$u_j(P_1, P_2, \cdots, P_M) = \sum_{a_1, \cdots, a_M} P_1(a_1) P_2(a_2) \cdots P_M(a_M) u_j(a_1, \cdots, a_M)$$

	A	B
A	1, 1	0, 0
B	0, 0	2, 2

(A, A)

(B, B)

Incomplete-information game and Bayesian NE

In a incomplete-information game:

The environment first draws $\theta = (\theta_1, \theta_2, \cdots, \theta_n) \sim p$ (known to all players)

↑ observation

Player i observes θ_i

Player i chooses p_i

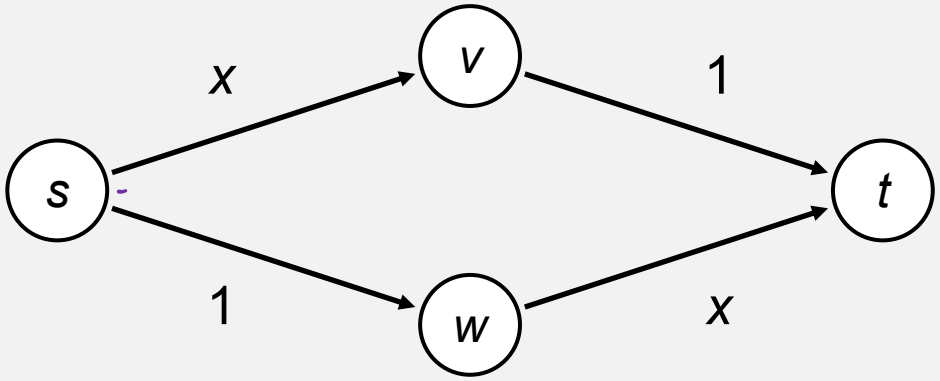
Bayesian NE

$$\mathop{\mathbb{E}}_{\theta \sim p}\left[u_i\left(p_i(\cdot|\theta_i), p_{-i}(\cdot|\theta_{-i}), \theta\right)\right] \geq \mathop{\mathbb{E}}_{\theta \sim p}\left[u_i\left(p_i'(\cdot|\theta_i), p_{-i}(\cdot|\theta_{-i}), \theta\right)\right] \quad \forall p_i'$$

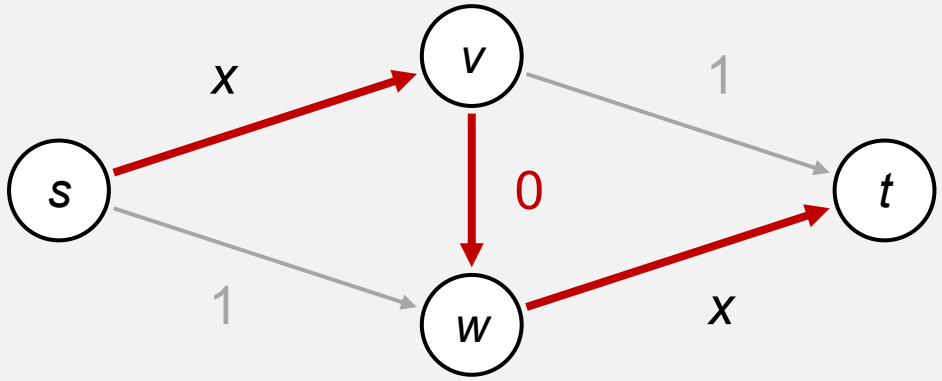
NE Example: Braess's paradox

1 unit of traffic goes from s to t, and every driver takes a fastest route.
Edge labels are travel times, where *x* = fraction of drivers using that edge.

Before



After: add a road v → w with travel time 0



NE Example: Second-Price Auction

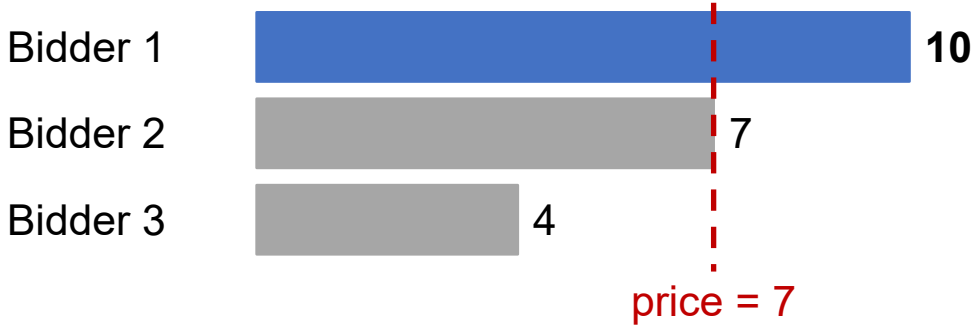
$b_i = v_i$ (NE)

Rules

- Each bidder i privately knows their value v_i
- Bidders submit sealed bids b_i simultaneously
- The highest bid wins
- The winner pays the second-highest bid

$$u_i(b_i, b_{-i}) = \begin{cases} v_i - \max_{j \neq i} b_j & \text{if } b_i > \max_{j \neq i} b_j \\ 0 & \text{otherwise} \end{cases}$$

Example (everyone bids truthfully)



Why bid truthfully?

Fix bidder i (value v, bid b) and let $B = \max_{j \neq i} b_j$ be the highest competing bid.

	Truthful b = v	**Overbid b > v**	**Underbid b < v**
B < v	Win: utility v − B	Same as truthful	May lose: utility 0
B > v	Lose: utility 0	May win: utility v − B < 0	Same as truthful

In every case, b = v is at least as good, so truthful bidding is a dominant strategy

Does NE Really Reflect the Players' Behaviors

- Human do not always act according to NE strategy. For example, in the second price auction, it is found that people tends to overbid
- Algorithmically, independent utility maximization algorithms do not always lead to NE
- Computing NE in a general-sum game is PPAD-hard
  - Existing algorithms requires $\left(M \sum_j A_j\right)^{O\left(\sum_j A_j\right)}$ time find an NE

$P \subseteq PPAD \subseteq NP$

Kagel, Harstad & Levin (1987), "Information Impact and Allocation Rules in Auctions with Affiliated Private Values: A Laboratory Study

Gradient Descent/Ascent in Two-Player Zero-Sum Games

Example. Gradient descent /ascent in two-player zero-sum games leads to *divergence* from the NE.

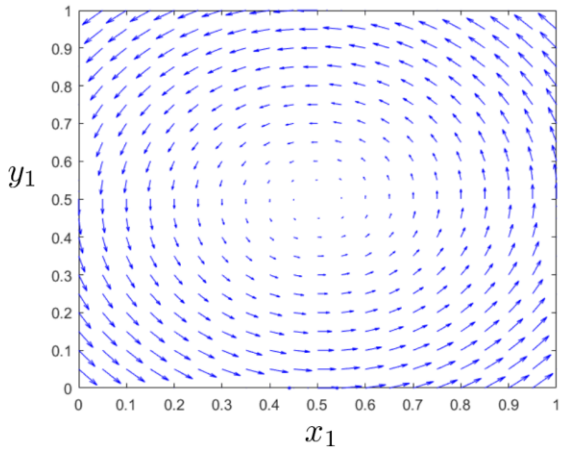
$$x_{t+1} = x_t + \eta \nabla_x f(x_t, y_t)$$

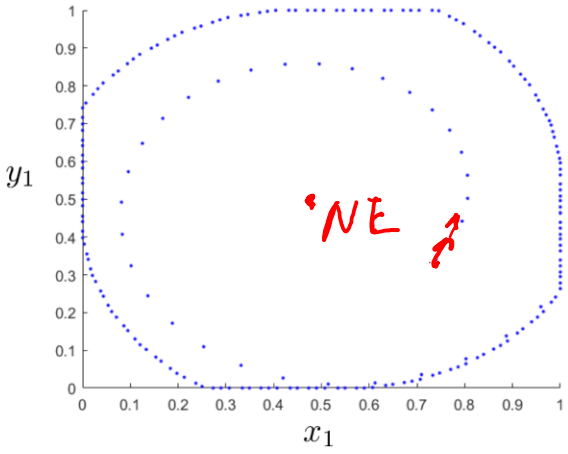
$$y_{t+1} = y_t - \eta \nabla_y f(x_t, y_t)$$

$\bar{x} = \sum_{t=1}^{T} x_t \qquad \bar{y} = \sum_{t=1}^{T} y_t$

$(\bar{x}, \bar{y}) \to NE.$

(x_t, y_t)





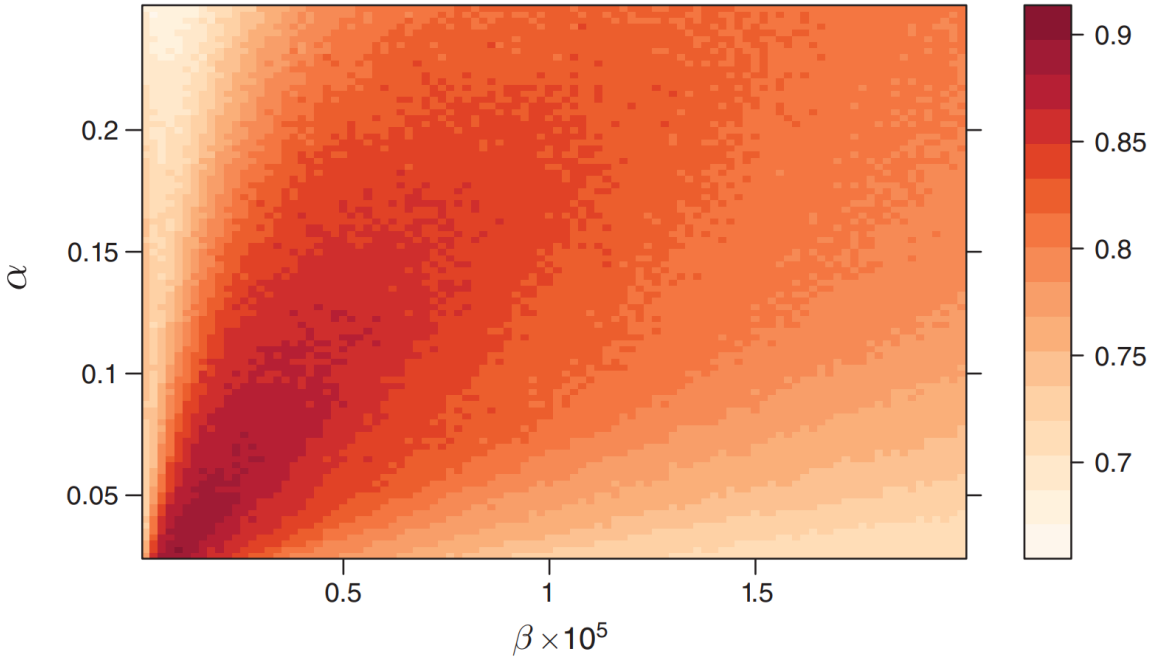
Independent Q-Learning in Pricing Game

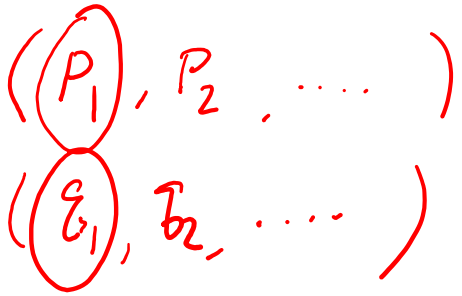
$$\Delta = \frac{\bar{R} - R^{\mathrm{NE}}}{R^{\mathrm{Monopoly}} - R^{\mathrm{NE}}}$$

R^{Monopoly}: profit in full cooperation

R^{NE}: profit in full competition
(i.e., equilibrium profit in one-shot game)

collusion





Calvano et al. Artificial Intelligence, Algorithmic Pricing, and Collusion. 2020.

Learning in Games

General questions:

$(a_{t,1}, a_{t,2}, \cdots, a_{t,m})$

- What does the empirical distribution of joint play converge to, or approach, when players interact using specific learning algorithms?
- What can a player using a specific algorithm guarantee about their average utility?

For example,

- If every player uses Q-learning
- If every player uses no-(swap)-regret algorithm (symmetric)
- If one player uses no-(swap)-regret algorithm and the other player knows it and optimizes over it

Correlated Equilibrium

and its connection to no-swap-regret dynamics

Correlated Equilibrium

Setting is as before

Additionally, there is "mediator" that will give suggested actions to every player.

Correlated Equilibrium. $\sigma \in \Delta(A_1 \times A_2 \times \cdots \times A_M)$ is a CE iff

$$\mathop{\mathbb{E}}_{a_{-j}} \left[u_j(a_j, a_{-j}) \middle| a_j \right] \geq \mathop{\mathbb{E}}_{a_{-j}} \left[u_j(a_j', a_{-j}) \middle| a_j \right] - \varepsilon$$

$$\| \qquad\qquad \geq$$

$$\forall a_j \in \text{supp}(\sigma) \quad \mathop{\mathbb{E}}_{a_{-j} \sim \sigma(\cdot|a_j)} \left[u_j(a_j, a_{-j}) \right] \qquad \mathop{\mathbb{E}}_{a_{-j} \sim \sigma(\cdot|a_j)} \left[u_j(a_j', a_{-j}) \right] - \varepsilon$$

↑ expected payoff if i'm recommended a_j

↑ expect payoff if we don't follow the recommendation a_j

Correlated Equilibrium Example: Traffic Light

Two drivers meet at an intersection (a Chicken game). A traffic light picks one of the three cells marked 1/3 at random, and shows each driver only their own light.

	Stop	**Go**
Stop	6, 6 prob. 1/3	2, 7 prob. 1/3
Go	7, 2 prob. 1/3	0, 0 prob. 0

If we are recommended "Stop":

$\Rightarrow Pr(\text{the other player is recommended "Stop"}) = \frac{1/3}{1/3+1/3} = 1/2$

$Pr(\text{_____ "Go"}) = 1/2$

$\Rightarrow \mathbb{E}[u_1(\text{Stop}, a_2) \mid \text{stop}] = \frac{6+2}{2} = 4$ (BR)

$\mathbb{E}[u_1(\text{go}, a_2) \mid \text{stop}] = \frac{7+0}{2} = 3.5$

If _____ "Go":

$\mathbb{E}[u_1(\text{Go}, a_2) \mid \text{Go}] = 7$ (BR)

$\mathbb{E}[u_1(\text{Stop}, a_2) \mid \text{Go}] = 6$

No-Swap-Regret Learning converges to Correlated Equilibria

Let M players repeated play a normal-form game with each other using no-swap-regret algorithms.

Let the action profile be $(a_{t,1}, \dots, a_{t,M})$ in round t.

Then their ~~average joint distribution~~ converges to a CE.

$$\sigma(a_{-j} \mid a_j) = \frac{\sum_t \mathbb{1}(a_{t,j} = a_j, a_{t,-j} = a_{-j})}{\sum_t \mathbb{1}(a_{t,j} = a_j)}$$

$$\sigma(a_1, a_2, \cdots, a_M) = \frac{1}{T} \sum_{t=1}^{T} \mathbb{1}(a_{t,1} = a_1, \cdots, a_{t,M} = a_M)$$

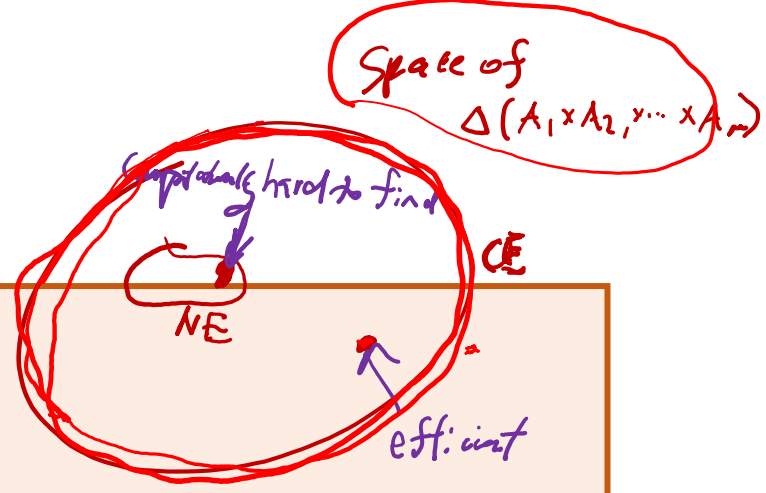
P.S. by swap regret, $\forall j, \forall a_j'$

$$\sum_{t=1}^{T} \mathbb{1}(a_{t,j} = a_j) \left(u_j(a_j', a_{t,-j}) - u_j(a_j, a_{t,-j}) \right) \le o(T)$$

$$\Rightarrow \sum_{t=1}^{T} \sum_{a_{-j}} \mathbb{1}(a_{t,j} = a_j, a_{t,-j} = a_{-j}) \left(u_j(a_j', a_{-j}) - u_j(a_j, a_{-j}) \right) \le o(T)$$

$$\Rightarrow \sum_{t=1}^{T} \mathbb{1}(a_{t,j} = a_j) \mathop{\mathbb{E}}_{a_{-j} \sim \sigma(\cdot \mid a_j)} \left(u_j(a_j', a_{-j}) - u_j(a_j, a_{-j}) \right) \le o(T)$$

Nash Equilibria ⊆ Correlated Equilibria



$\sigma \in \Delta(\mathcal{A}_1 \times \cdots \times \mathcal{A}_M)$ is a Correlated Equilibrium iff

$\forall j \quad \forall\, a_j, a_j' \in \mathcal{A}_j$ with $a_j \in \text{support}(\sigma)$

$$\mathbb{E}_{a_{-j} \sim \sigma(\cdot|a_j)}\left[u_j(a_j, a_{-j})\right] \geq \mathbb{E}_{a_{-j} \sim \sigma(\cdot|a_j)}\left[u_j(a_j', a_{-j})\right]$$

$\sigma(a_1, a_2, \cdots, a_M) = P_1(a_1)\, P_2(a_2) \cdots P_M(a_M)$

$(p_1, p_2, \cdots, p_M) \in \Delta(\mathcal{A}_1) \times \cdots \times \Delta(\mathcal{A}_M)$ is a Nash Equilibrium iff

$\forall j \quad \forall\, p_j' \in \Delta(\mathcal{A}_j) \qquad u_j(p_j, p_{-j}) \geq u_j(p_j', p_{-j})$

$u_j(p_j, p_{-j}) = \mathbb{E}_{a_j \sim p_j} \mathbb{E}_{a_{-j} \sim p_{-j}} \left(u_j(a_j, a_{-j})\right)$

$P_1 = (1, 0)$, $P_2 = (1, 0) \Rightarrow \sigma =$ 

Coarse Correlated Equilibrium

and its connection to no-regret dynamics

Coarse Correlated Equilibrium (CCE)

σ is a CE $\Rightarrow$ σ is a CCE

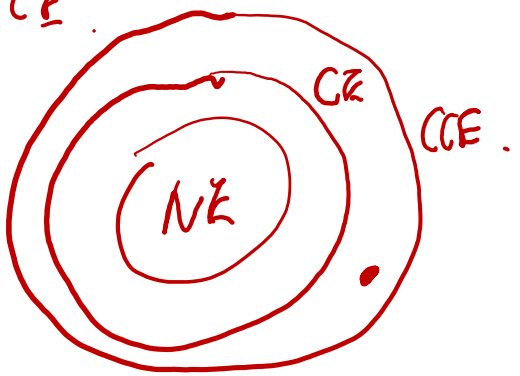
CE: $\mathbb{E}_{a_{-j}\sim\sigma(\cdot|a_j)}[u_j(a_j, a_{-j})] \geq \mathbb{E}_{a_{-j}\sim\sigma(\cdot|a_j)}[u_j(a'_j, a_{-j})]$

CCE: $\mathbb{E}_{a\sim\sigma}[u_j(a_j, a_{-j})] \geq \mathbb{E}_{a_{-j}\sim\sigma}[u_j(a'_j, a_{-j})]$ $\forall a'_j$

$(a_1, \cdots, a_m)$

Check : For every player. follow the recommendation is no worse than choosing a fixed action

space of deviation in CE $\supseteq$ space of deviation CCE.



Coarse Correlated Equilibrium Example

	Rock	Paper	Scissor
Rock	0,0 prob. 0	-1, 1 prob. 1/6	1, -1 prob. 1/6
Paper	1, -1 prob. 1/6	0, 0 prob. 0	-1, 1 prob. 1/6
Scissor	-1, 1 prob. 1/6	1, -1 prob. 1/6	0, 0 prob. 0

payoff if follow recommendation = 0

payoff if one player deviates to a fixed action = 0

⇓

σ is a CCE

---

for row-player

If we are recommended "rock"

⇒ the other player uses paper & scissor w.p. $\frac{1}{2}$

payoff if follow recommendation = 0

if deviate to scissor $= \frac{1+0}{2} = \frac{1}{2}$

⇓

σ is not a CCE

No-Regret Learning converges to CCE

Nash Equilibria ⊆ Correlated Equilibria ⊆ Coarse Correlated Equilibria

Summary

- NE $\subseteq$ CE $\subseteq$ CCE
- If every player runs a no-swap-regret algorithm, their average profile is an approximate CE
- If every player runs a no-regret algorithm, their average profile is an approximate CCE

Stackelberg Equilibrium

and its connection to no-(swap-)regret + arbitrary other-player dynamics

Ordering Matters in General-Sum Games

	A	**B**
A	**(2, 1)**	(0, 0)
B	(0, 0)	**(1, 2)**

Stackelberg Equilibrium

[assume player 1 is the first mover / leader]

$(p, q) \in \Delta(A_1) \times \Delta(A_2)$ is a Stackelberg equilibrium if

$BR(p) = \arg\max_q u_2(p, q)$

① $q = BR(p)$

(with some tie-breaking rule for P2)

② $p = \arg\max_p u_1(p, BR(p))$

$$\begin{cases} x^* = \arg\max_x \min_y f(x, y) \\ y^* = \min_y f(x^*, y) \end{cases}$$

Repeated Game between no-(swap-)regret algorithm & another player

(follower)

- No-(swap-)regret algorithm ≈ Best-responding player as if they know the other player's strategy
- If we are the other player, how can we optimize our strategy?

(leader)

Learner’s Viewpoint

Recall: Two-Player Zero-Sum Games

- A no-regret algorithm guarantees the learner to achieve a value **no worse than the game value** by repeated play (even when the other player uses an arbitrary strategy)
- In other words: even if the online learning algorithm is a first mover, its long-term payoff is no worse than being the second mover (second mover is usually more advantageous as it has more information to make decisions)

No-Regret in General-Sum Games

Let $(y_1, \ldots, y_T)$ be a no-regret learner's plays, and $(x_1, \ldots, x_T)$ be an arbitrary player's plays.

Then

$$\frac{1}{T}\sum_{t=1}^{T} u_2(x_t, y_t) \geq \max_y u_2(\overline{x}, y) - o(1)$$

where $\overline{x} = \frac{1}{T}\sum_{t=1}^{T} x_t$.

$$= \max_y \frac{1}{T}\sum_{t=1}^{T} u_2(x_t, y)$$

No-Swap-Regret in General-Sum Games

Let $(y_1, \dots, y_T)$ be a no-swap-regret learner's plays, and $(x_1, \dots, x_T)$ be an arbitrary player's plays.

Then

$$\frac{1}{T}\sum_{t=1}^{T} u_2(x_t, y_t) \geq \sum_{y} c_y \max_{y'} u_2(\bar{x}_y, y') - o(1) \geq \max_{y} u_2(\bar{x}, y) - o(1)$$

where $c_y = \frac{1}{T}\sum_{t=1}^{T} \mathbb{I}\{y_t = y\}$ and $\bar{x}_y = \frac{\sum_{t=1}^{T} x_t \mathbb{I}\{y_t = y\}}{\sum_{t=1}^{T} \mathbb{I}\{y_t = y\}}$.

$\bar{x} = \sum_y c_y \bar{x}_y$

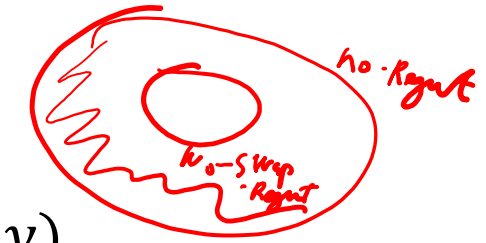
Optimizer’s Viewpoint

How to play against a no-(swap-)regret player?

- Regard yourself as the first-mover (a.k.a. leader) in the Stackelberg game, and view the other player as the second-mover (a.k.a. follower) who will best respond to you.

What can the optimizer achieve?

$$V_{\text{stack}} := \max_{x} \max_{y \in \text{BR}(x)} u_1(x, y) \quad \text{where } \text{BR}(x) = \text{argmax}_y \, u_2(x, y)$$



Learner (Player 2)'s algorithm	What can Optimizer (Player 1) achieve?	Can Optimizer (Player 1) do better?
No-Regret	$\frac{1}{T}\sum_{t=1}^{T} u_1(x_t, y_t) \geq V_{\text{stack}} - o(1)$ ①	Sometimes ③
No-Swap-Regret		No $\frac{1}{T}\sum_{t=1}^{T} u_1(x_t, y_t) \leq V_{\text{stack}} + o(1)$ ②

Yuan Deng, Jon Schneider, Balusubramanian Sivan. Strategizing against No-regret Learners. NeurIPS 2019.

Optimizer can achieve $\geq V_{\text{stack}}$ against any no-regret ①

Optimizer can achieve $\leq V_{\text{stack}}$ against any no-swap-regret ②

Consider a fixed Learner's action y.

Define $\bar{x}_y$ as the Optimizer's action distribution when Learner chooses y:

$$\bar{x}_y = \frac{\sum_{t=1}^{T} \mathbb{I}[y_t = y] x_t}{\sum_{t=1}^{T} \mathbb{I}[y_t = y]}, \qquad c_y = \frac{1}{T} \sum_{t=1}^{T} \mathbb{I}[y_t = y].$$

$$\begin{aligned}
&\frac{1}{T} \sum_{t=1}^{T} \big(u_2(x_t, \sigma(y_t)) - u_2(x_t, y_t)\big) \\
&\quad = \sum_{y \in \mathcal{A}_2} \frac{1}{T} \sum_{t=1}^{T} \mathbb{I}[y_t = y] \big(u_2(x_t, \sigma(y)) - u_2(x_t, y)\big) \\
&\quad = \sum_{y \in \mathcal{A}_2} c_y \big(u_2(\bar{x}_y, \sigma(y)) - u_2(\bar{x}_y, y)\big).
\end{aligned}$$

Optimizer can achieve $\leq V_{\text{stack}}$ against any no-swap-regret ②

As Player 2 has no-swap-regret, for $y \notin \text{BR}(\bar{x}_y)$, we have $c_y = o(1)$.

Optimizer utility:

$$\begin{aligned}
\frac{1}{T}\sum_{t=1}^{T} u_1(x_t, y_t) &= \sum_{y \in \mathcal{A}_2} c_y u_1(\bar{x}_y, y) \\
&= \sum_{y:\, y \in \text{BR}(\bar{x}_y)} c_y u_1(\bar{x}_y, y) + \sum_{y:\, y \notin \text{BR}(\bar{x}_y)} c_y u_1(\bar{x}_y, y) \\
&\leq \sum_{y:\, y \in \text{BR}(\bar{x}_y)} c_y V_{\text{stack}} + \sum_{y:\, y \notin \text{BR}(\bar{x}_y)} c_y \cdot 1 \\
&= V_{\text{stack}} + o(1).
\end{aligned}$$

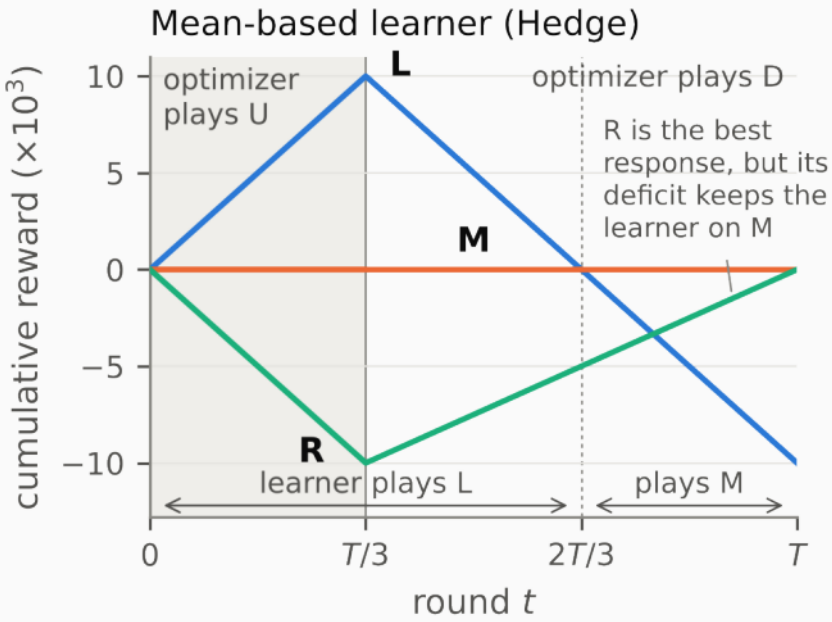
Optimizer may achieve $\geq V_{\text{stack}} + \Omega(1)$ against some no-regret

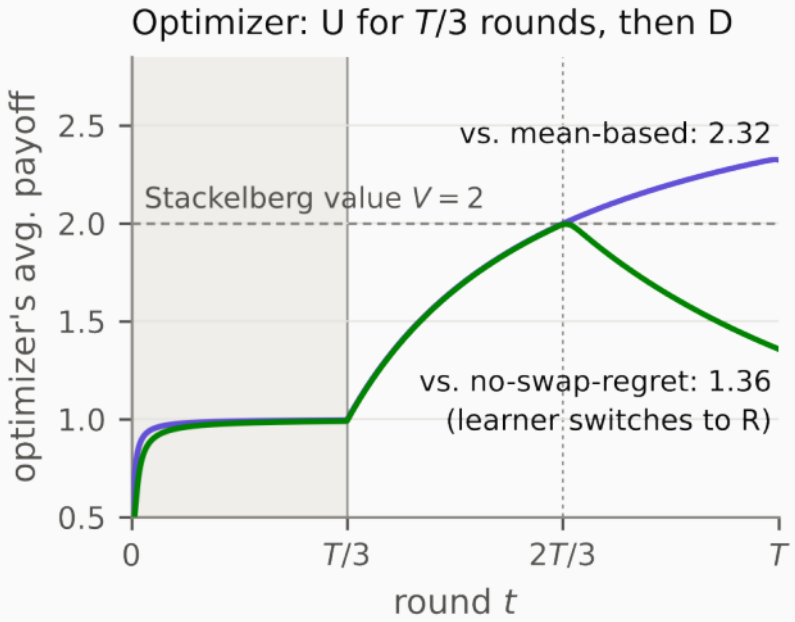
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	L	M	R
U	$(1, 1)$	$(0, 0)$	$(0, -1)$
D	$(3, -1)$	$(3, 0)$	$(0, \frac{1}{2})$

optimizer's payoff first; $V_{\text{Stack}} = 2$

Intuition. Hedge follows the best *cumulative* reward: L's lead and R's deficit keep it on the wrong action. No swap regret forbids this.





From Monday's slides

Bayesian Persuasion

An imperfect information Stackelberg game

Example: Professor recommending students

- Professor recommending students to a school
- There are two types of students: **strong** and **weak**
- Professor hopes every students can be admitted; school hopes to admit strong students and reject weak student
- Professor knows the type of every student; school only knows the distribution

School action

Entry: professor payoff, school payoff

	Admit	Reject
Strong	1,1	0,0
Weak	1,-1	0,0

Student type

Pr(Strong) = 0.3, Pr(Weak) = 0.7

School (Follower)'s Strategy: Best Response

$P(\text{strong} \mid \text{rec}) \propto P(\text{strong})\, P(\text{rec} \mid \text{strong})$ — $P(\text{strong})$: Professor strategy

Assume that after the school receives $a \in \{\text{recommend}, \text{not recommend}\}$

The school can calcute the posterior distribution for the type of the student

$P(\text{type} \mid a)$

$$\begin{cases} P(\text{strong} \mid \text{rec}) = \xi \\ P(\text{weak} \mid \text{rec}) = 1-\xi \\ P(\text{strong} \mid \text{not rec}) = r \\ P(\text{weak} \mid \text{not rec}) = 1-r \end{cases}$$

$2\xi - 1 \geq 0 \Rightarrow \xi \geq \frac{1}{2}$

Entry: professor payoff, school payoff

	Admit	Reject
Strong	1,1	0,0
Weak	1,-1	0,0

Pr(Strong) = 0.3, Pr(Weak) = 0.7

$$\Rightarrow \begin{cases} E[\text{payoff}(\text{admit}) \mid \text{rec}] = \xi \cdot 1 + (1-\xi)(-1) = 2\xi - 1 \\ E[\text{payoff}(\text{reject}) \mid \text{rec}] = 0 \end{cases}$$

$E[\quad \mid \text{not rec}] = 2r - 1$

$E[\quad \mid \text{not rec}] = 0$

Professor (Leader)'s strategy

① Always recommend ⇒ Best response = always reject ⇒ Professor payoff = 0

② Truthful recommendation ⇒ Best response = follow the recommendation ⇒ Professor payoff = 0.3

Entry: professor payoff, school payoff

	Admit	Reject
Strong	1,1	0,0
Weak	1,-1	0,0

Pr(Strong) = 0.3, Pr(Weak) = 0.7

Professor (Leader)'s strategy

$0.3 \times 1 + 0.7 \times \frac{3}{7} = 0.6$

Professor's persuasion strategy

	Positive recommendation	Negative recommendation
Strong	1	0
Weak	$3/7 - \varepsilon$	$4/7 + \varepsilon$

Entry: professor payoff, school payoff

	Admit	Reject
Strong	1,1	0,0
Weak	1,-1	0,0

Pr(Strong) = 0.3, Pr(Weak) = 0.7

$$\begin{cases} P(\text{strong} \mid \text{rec}) \propto P(\text{strong})\,P(\text{rec} \mid \text{strong}) = 0.3 \cdot 1 = 0.3 \\ P(\text{weak} \mid \text{rec}) \propto P(\text{weak})\,P(\text{rec} \mid \text{weak}) = 0.7 \cdot \frac{3}{7} = 0.3 \end{cases}$$

$\propto$: $\frac{1}{2}$ $\quad$ $=$: $\frac{1}{2}$

Bayesian Persuasion Game

- Leader observes context C drawn from distribution $\mathcal{D}$ ($\mathcal{D}$ is public but C is private to the Leader)
- Leader chooses a_L based on C
- Follower chooses a_F after observing a_L
- Leader gains $u_L(C, a_L, a_F)$ and $u_F(C, a_L, a_F)$